

BDA Assignment-1

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① Explain detail about HDFS?

1. HDFS stands for Hadoop Distributed File System.
2. Storage system of Apache Hadoop.
3. It is designed to store very large files.
4. It works in a distributed environment.
5. HDFS follows a master-slave architecture.

Architecture:-

- NameNode is the master node.
- It stores metadata.
- It does not store actual data.
- Data nodes are slave nodes.
- Data nodes send heartbeat signals to the NameNode.

Working:-

- Files are divided into large blocks.
- Each block is stored in different data nodes.
- Blocks are replicated.
- Replication provides fault tolerance.
- If one data node fails, data can be accessed by another copy.

Limitations:-

- Not suitable for small files
- Not suitable for real-time processing
- Follows write once read ~~many~~ many nodes.

* HDFS is a reliable and scalable distributed file system used for storing and managing big data efficiently in hadoop environments.